Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

MACHINE LEARNING- (25SC2107E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No- 05	Team No-	Project Title- Rainfall-Based Yield Forecasting Using Gradient-Boosted Trees with Bayesian Hyperparameter Search

1. Abstract

Agricultural crop yield is significantly influenced by rainfall patterns and other environmental factors, making accurate yield forecasting important for agricultural planning and resource management. This project presents a machine learning-based **Rainfall-Based Yield Forecasting System** that predicts crop yield using historical rainfall and relevant agricultural features. The proposed system follows the complete machine learning lifecycle, including data preprocessing, feature selection, model training, validation, evaluation, and prediction.

A **Gradient-Boosted Tree** model is employed to capture complex non-linear relationships between rainfall conditions and crop yield. To improve the model's predictive performance, **Bayesian Hyperparameter Search** is applied to efficiently identify optimal hyperparameters. The model is evaluated using **cross-validation** and regression metrics such as **Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 score**. Baseline regression models such as Linear Regression, Ridge Regression, or Lasso Regression can also be used for performance comparison. Feature importance analysis is performed to identify the rainfall and agricultural factors that contribute most to yield prediction.

The proposed system aims to provide accurate and data-driven crop yield estimates that can assist farmers and agricultural planners in making informed decisions regarding crop planning, irrigation, and resource allocation. The project demonstrates the application of tree-based supervised learning, model evaluation, and Bayesian hyperparameter optimization in a practical agricultural forecasting problem.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520030459	B. Koushika	
2520030005	M. Bhavya	

Guide Name and Signature (remarks if any) _____